

PUT AND GET 2/1/72 (PG,11)

/FNN (BPC)

CTEM=202

ATEM=112

PG1=ATEM

PG2=ATEM+1

PG3=ATEM+2

PG4=ATEM+3

PG6=ATEM+4

PG7=ATEM+5

PG8=ATEM+6

PG9=ATEM+7

PG10=ATEM+10

SGI=IOT 5300

NNIF=IOT 6300

/READ ROUTINE FOR PUT AND GET
 /GETS PIECE OF ITEM OR MAP INTO CORE
 /IO=ADDR OF DESIRED WORD, REL TO BEG OF INFO
 /USES F,G,H
 /RETURNS CORE ADDR OF DESIRED WORD IN IO

RDM,	DAC I (PG7)	
	STF 6	/READ, MAP
	SZF I 3	
	JMP COREM	
	JMP RD	
VRL,	LAW VRR	
RDI,	DAC I (PG7)	
	CLF 6	/READ, ITEM
	SZF I 2	
	JMP COREI	
RD,	DIO I (PG6)	
	LAW 3	
	SZF 6	
	ADD (5)	
	ADD F	
	DAC G	/REL ORG OF CUR PIECE
	LAW I 2	
	SZF 6	
	LAW 2	
	SUB I G	
	ADD I (PG6)	/CAL ADR REL TO BEG OF PIECE
	SPA	
	JMP WR1A	/MINUS= NOT IN CORE
	DAC H	
	IDX G	
	IDX G	
	LAC I 0	/LTH OF PIECE
	SUB F	
	SPC	
	JMP WR1	/BEYOND END
	LAW I 1	/WORD DESIRED ALREADY IN CORE
	ADD 0	
	DAC C	
	LAC I G	/CORE ADD OF PIECE
	ADD H	
	LIA	/PUT CORE ADD OF WORD IN IO
	JMP PDF3	
COREM,	LAC I (PG10)	
	JMP CORE	
COREI,	LAC I (PG9)	
CORE,	AAI	
	LIA	
	JMP PDF3	

```

UCRD,   DAC I (PG7)
        STF 6           /UNCOND READ, MAP
        DIO I (PG6)
WR1,    LIO I (PG10)
        SZF 6
        JMP RD1
        LIO I (PG9)     /ITEM ADD
        SZF I 5
        JMP RD1
        CLF 5
WR2,    LAC CIORIN      /SET UP WRITE COMMAND
        DIP I F
        LAW 4
        ADD F
        DAC H
        LAC CIORIN
        DIP I H
        JSP SGCM

RD1,    LAC CIORIN
        DIP I F
RD1A,   LAW 3
        ADD F
        SZF 6
        ADD (5)
        DAC H
        IQA
        DAC 6
        CLA           /SET UP READ COMMAND
        DIP I G
        LAW I 2
        SZF 6
        LAW 2
        ADD I (PG6)
        DAC I H
        JSP SGCM

```

```

PDF1,    LAW 4
          ADD F
          SZF 6
          ADD (5)
          DAC G
          LIO I G      /RET BEG OF BUF
PDF3,    LAC I (PG7)
          DAC F
          JMP HEXIT
UCRW,    DAC I (PG7)
          CLF 6
          SZF I 2
          JMP COREI
          DIO I (PG6)
WR1A,    SZF 6
          JMP WR1
          LAW I 1
          ADD I (PG6)
          SZM
          JMP WR1
          LAC F
          IDA
          DAC G
          LAC I F
CANDIN,  AND (7777)
          ADD I G
          SUB B16
          ADD I (PG6)
          LIA
          JMP RDF3
UCRI,    DAC I (PG7)
          CLF 6
          DIO I (PG6)
          CLA
          DIP I F
          LIO I (PG9)
          JMP RD1A

```

SGCH, DAP H
 LAC I (PG8)
 SAP 1S
 AND (3)
 ADD (XCT SGTB)
 DAC G
 LAC F
 SZF 6
 ADD (7)
 XCT G
 JMP FSTER1
 JMP HEXIT

SGTB, SGI+1
 SGI+3
 SGI+5

FSTER1, LAC I (ERCODE) /WRITE ERROR
 SAD (40000) /ILL SPEC ERROR
 JMP HEXIT

FSTER3, DIO I (ERCODE+1)
 LIO I (PG8)
 LAC I (USERPG)
 DAC I (TRAPPG)
 IDA
 RIR 1S
 SPI I
 LAW IOPTSU
 JMP GO

/BEGIN ITEM

BG1C, LAW 10
 DAC I (PG8)
 LAC I (USERID)
 DAC I (PG9)
 LIO I (USERAC)
 DIO I (PG10)
 SEF I 1
 JMP MPF
 JSP GSET
 LIO B16
 JSP UCRW
 JMP PRT

BG12, JSP PTFX
 LAC H
 DAC A
 CLI
 JSP RDI
 DIO B
 DIO E
 IDX B
 LAW 1760
 AND I C
 RAR 4S
 DAC I E
 SUB B16
 DAC I (CTEM) /REL PTR FROM TREE REG TO TREE
 LAW 17
 AND I C
 ADD I E
 IDA
 DAC I A
 LAC B1
 DAC I B
 LAW 1
 DAC I (PG4)
 LAW 7
 DAC I (PG3)

BGA, LIA
 JSP RDM
 DIO B
 LAC I B
 AND CJ
 SEA
 JMP BGS
 IDX I E
 IDX I A
 STF 5
 IDX I (PG4)
 LIA
 JSP RDI
 DIO B
 LIO I (CTEM)
 DIO I B

LAW 3
ADD I (PG3)
DAC I (PG3)
JMP BGA

RGB,
IDY I (PG4)
ADD I (CTEM)
DAC A

BGC,
LID I (PG4)
STF 5
JSP RDI
DIO B
CLC
DAC I B
IDY I (PG4)
SAS A
JMP BGC
LIA
STF 5
JSP RDI
DIO B
LAC ML4
DAC I B
STF 5
SZF I 1
JMP PRT
CLI
JSP RDM
DIO C
JMP GTP3

/EOM

/GET A FIELD

```

GETX,      CLL+UCML
GETC,      LAH GET1
GSET,      DAP H
           LIF+USCF
           DIO I (PG8)      /SAVE FLAGS FROM IOT
           RIP 6S
           SPI
           STF 1

           LAC I (USERAC) /GET PARAMETERS
           DAC A
           LAC I A
           DAC I (PG1)      /SAVE INFO ADD OR ERRORS AREA
           IDX A
           LAC I A
           DAC D
           IDX A
           LAC I A      /ITEM ADD
           SPA
           CMA+USTF 2      /IF DRUM ADD, COMP AND SET FLAG 2
           DAC I (PG9)
           IDX A
           LAC I A      /MAP ADD
           SPA
           CMA+USTF 3      /IF DRUM ADD, COMP AND SET FLAG 3
           DAC I (PG10)
           IDX A
           LAC I A
           DAC I (PG2)
           IDX A
           LAC I A
           DAC F      /SAVE BUFFER ADDRESS
           SZF 2
           LAC I F
           SPA
           STF 5
           JMP HEXIT
GET1,      RIL 1S      /INFO PRES FLAG
           SPI
           JMP GTP1
           CLI+UCLF 5
           SZF I 3
           LIO I (PG10)      /MAP IN CORE
           SZF 3
           JSP UGRD
MPF,      DIO C
           SZF I 2
           JMP NIT
           LAH 1
           ADD F
           DAC H
           LAH 17
           AND I C
           ADD 816
           DAC I H

```

LIO 816
JSP UCR1

GTP4,
 LAC F
 ADD (5)
 DAC E
 LAC I E
 DAC E
 LAC I F
 AND (7777)
 DAC A
 LAN 17
 AND I C
 ADD B16
 CHARLC,
 SUB I A
 ADD E
 SMA
 STF 4

GTP,
 LAC I (PG8)
 RAR 4S
 SPA

JMF BG12
 GTP3,
 LAN I 1777

AND I C
 RAL 8S
 SUB D

/MAX FIELD ADD

SPA
 JMP GTN1

/FIELD ADDR TOO HIGH

LIO D

JSP RDM

DIO B

/CORE ADDR= WD 0 OF FIELD MAP

LIO I B

DIO B

SPI

JMP FCD

/SEC INFO

DZM I (PG3)

FTB,
 IDX D
 DAC C
 IDX C
 LIO D
 JSP RDM
 DIO D
 LAN I 7600
 AND I C
 DAC D
 ADD I (PG3)
 AND (17777)
 DAC A
 LIO C
 JSP RDM
 DIO C
 LIO A
 LAC I C
 SPA
 SZF 4
 JMP VRL
 JSP UGRW

/JSP RDI

```

VRR,    LAC D
        AND (760000)
        IAI
        DAC E
        LIO B
        RIL 16
        SPI I
        JMP GTE3
        LIO I 0
        SPI
        JMP FTA      /FIXED LENGTH INFO
        CLI
        JSP RDI
        DIO D
        SZF 1
        JMP FPV
        LAC I 0
        ADD I E
        DAC E
        AND (177777)
        LIA
        LAW VRT
        SZF 4
        JMP RDI
        JSP UCRW
VRT,    LAC E
        AND ML2
        IAI
        LIO I 0

FTA,    SZF 1
        JMP FPF
        DIO I (USERIO)
        DAC I (USERAC)
PRT,    CLA
        SZF 5
        CMA
        SZF 2
        DIP I F
        LAC I (PG8)
        RAR 13
        SMA
        JMP R1
        JMP R2

```

```
GTP1,  CLI
        JSP RDM
        DIO C
        SZF 2
        JMP GTP4

NIT,   LAW 17
        AND I C
        ADD I (PG9)
        DAC I (PG9)
        JMP GTP

GTE1,  DAC I (USERAC)
        LAC I (PG1)
        JMP FTE1

GTE2,  SZF 1
        JMP FPE
        DAC I (USERAC)
        JMP FTE2

GTE3,  SZF 1
        JMP PRT
        DAC I (USERAC)
        LIO I C
        DIO I (USERIO)

FTE3,  IOX I (PG1)
FTE2,  IOX I (PG1)
FTE1,  SZF 1
        C16RET*1
        DAC A
        LAC I A
        JMP GO
```

```

FCD,      LAN CTEM
          DAC A
          LAC D
          DAC I A

CHARLO,   IDX A
FCG,      SAC (CTEM+6)
          C16RET+1      /TOO LOW= PUNT
          CLA
          RCR 8S
          RAL 8S      /POINTER TO NEXT LEVEL
          DAC I A      /PUSH
          LIA
          JSP RDM
          DIO H
          LIO I H
          SPI          /TREE BASE?
          JMP FCG      /NO= GO ON DOWN TREE

FRK,      IDX I A      /TREE POINTER
          LIO I A
          JSP RDM
          DIO H
          LAN 177
          AND I H
          DAC I (PG3)  /REL TREE LOC
          LIA
          JSP RDI
          DIO D
          LAC I D
          ADD I (PG3)
          DAC I (PG4)  /NEXT LEVEL LOC
          LAN I 1
          ADD A

```

```

FCC,   DAC C           /POP 1 LEVEL
        LIO I (PG4)
        JSP RDI
        DIO D           /NEXT LEVEL POINTER
        LIO I D
        DIO D
        LIO I (PG3)
        JSP RDI
        DIO B           /THIS LEVEL POINTER
        LIO I C
        JSP RDM
        DIO E           /MAP POINTER
        LAC I (PG2)
        DAC H
        LAC (77400)
        AND I E
        DAC E
        MUL I W         /SUBSCRIPT
        SQL 95
        ADD I B         /ADD POINTER
        ADD I (PG3)
        DAC I (PG3)     /NEXT LEVEL
        SUB D
        SUB I (PG4)
        SMA             /TOO HIGH?
        JMP GTE2        /ERROR 2= PUT MAYBE
        LAC E
        RAR BS
        ADD I (PG3)
        DAC I (PG4)     /NEXT LEVEL FOR CHECKING
        IOX I (PG2)
        LAH I 1
        ADD C
        DAC C
        SAS (CTEM-1)
        JMP FCC

FCF,   LIO I (CTEM)
        DIO D
        JSP RDM
        DIO B
        LIO I B
        DIO B
        JMP FTB

```

FPV,

LAW 1
ADD D
DAC B
LIO (PG1)
JSP TRACEI
LAC A
DAC C
LAC I B
DAC I E
STF 5

FV1,

IDC
DAC I B
AND (177777)
ADD I D
LIA+UCLF 1
JSP RU1
LAC I B
AND ML2
IAI
DAC E
LCH I C
DCH E
SAD (74)
STF 1
JSP PTFX
LAC E
AND ML2
SAD B1
IOX I 4
LAC I B
STF 5
SEF I 1
JMP FV1
JMP PRT

PRT1,


```

FPF,      LIA          /FIXED LENGTH FIELD
          SZL I        /SKIP CONVERSION IF LINK SET
          JSP PTCON
          DIO B
          LIO (PG1)
          JSP TRACEI
          RIP 5S       /PROGRAM BUM BECAUSE TRACE MASKS OUT EXTEND BITS
          SZL I        /SKIP CONVERSION IF LINK SET
          JSP PTCON
          DIO A

          LAH 777
          AND I C      /3RD WORD OF MAP
          SZL
          JMP FP2      /SKIP CALC BYTES IF LINK SET
          CLI
          ROP 5S
          RIL 6S
          DIO C
          MUL C18,
          SHP
          ADD C
          CHA+UCHI+USWP
          DIV C6
          6
          DAC C

          LCH I A
          DCH I H
          ISP C
          JMP ,=3
          JMP PRT1

FP2,      CHA
          JMP FP1

PTCON,   DAP H
          CLA+USWP
          ROL 6S
          DAC G
          CLA
          DIV C6
          18,
          LIO G
          RIP 4S
          ROR 2S
          JMP HEXIT
C18,,

```

/MAKE SPACE IN TREE

```

FPE,      SZA
          C16RET+1      /ERROR IF MORE THAN ONE OFF
          LAR CTEM
          SAS C
          C16RET+1      /ERROR IF NOT AT TOP LEVEL
          LIO I A
          JSP RDM
          DIO H
          LAR 177
          AND I H
          DAC B          /TREE POINTER LOCATION
          DAC D
          LAR E
          RAR 85
          DAC E          /ADJUST GROUP LENGTH
          LAR 2
          JMP PUB

```

```

PUC,      LIO B          /INCREMENT TREE POINTER
          JSP RDI
          DIO H
          LAR I H
          ADD E
          DAC I H
          STF 5
          LAR 3

```

```

CHARLA,PUB, ADD I A      /NEXT MAP FIELD
          DAC I A
          IDX R          /NEXT ITEM WORD
          LIO I A
          JSP RDM
          DIO H
          LAR I H
          AND CJ
          SZA I          /ANOTHER TREE POINTER
          JMP PUC        /YES
          LAR CTEM
          SUB A
          DAC A          /- NO OF LEVELS

```

```

PUD,      LIO D          /GO DOWN WITH SUBSCRIPTS 0
          JSP RDI
          DIO H
          LAR I H
          ADD D
          DAC D
          ISP A
          JMP PUD
          LIO I (CTEM+1)
          JSP RDM
          DIO A
          LAR (77400)
          AND I A
          RAR 85
          DAC A          /LTH OF GROUP ABOVE LEVEL

```

CHLNUM,


```

PUE,      LIO I (PG4)
          JSP RDI
          DIO G
          LAC I G
          ADD E
          DAC I G      /INC POINTER OF ABOVE LEVEL
          STF 5
          LAC I (PG4)
          SAD D        /LOWEST LEVEL
          JMP PUG      /YES, DONE
          ADD A        /NO, GO TO NEXT POINTER
          DAC I (PG4)
          JMP PUE

```

```

PUG,      LIO I (CTEM)
          JSP RDM
          DIO C
          LIO I C
          RIL 25
          DIO C
          LAC I (PG4)

```

```

PUH,      SPI I      /LOWEST LEVEL IN TREE
          SAD I (PG3) /OR LAST PTR ON LEVEL
          JMP PUK      /YES

```

```

          ADD E
          DAC I (PG4) /GO TO NEXT POINTER
          LIA
          JSP RDI
          DIO H
          LAC I H
          ADD E
          DAC I H
          STF 5
          LAC I (PG4)
          JMP PUH

```

```

PUK,      JSP PTFX
          LAC I H
          ADD E
          DAC I H      /INC WD CT
          CLI
          JSP RDI
          DIO A
          DIO B
          IDX B
          LAC I A
          ADD E
          DAC I A      /INC BEG OF VAR
          STF 5
          ADD I B
          AND (7777)   /REL INS LOC

```

```

PUJ,      DAC B
          SUB E
          DAC A
          LIO A
          JSP RDI
          DIO D
          LIO I D
          DIO D
          LIO B
          JSP RDI
          DIO H
          LIO D
          DIO I H
          STF 5
          LAC A
          SAD I (PG3)      /STOP AT INS PT
          JMP PUIC          /DONE
          LAW I 1
          ADD B
          JMP PUJ

PUIC,     LIO C
          SPI I
          JMP PUID          /LOWEST LEVEL IN TREE?
                               /NO

PUIB,     CLC
PUIA,     DAC I H
          STF 5
          LAW I 1
          ADD B
          DAC B
          SAD I (PG3)
          JMP FCF
          LIA
          JSP RDI
          DIO H
          JMP PUIB

PUID,     LAC I H
          SUB E
          JMP PUIA

PTFX,     DAP A
          SEF 2
          JMP PTFX2
          CLI
          JSP RDM
          DIO H
          LAN 17
          AND I H
          SUB I (PG9)
          CMA

PFX,      DAP H
          JMP AEXIT

PTFX2,    LAC I F
          JMP PFX

```

START

FORM 8510

PRINTED IN U.S.A.